INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: B.Tech, Stream: ALL (Group- A)

Term - I, First Semester Examination, August - 2025

Subject Code: BSCPH101

Subject Name: Physics

Full Marks: 30 Duration: 1 Hour

Part - A

Attempt any 5 questions

Each question carries 2 marks (2×5)

1.	Define vector field. Give an example.	2
2.	A simple harmonic oscillator is characterized by y = cos ωt. Calculate the displacement at kinetic energy is equal to its potential energy.	2
3.	What are eigen function and eigen value?	2
4.	Draw the intensity distribution of Young's double slit experiment.	2
5.	Define source and sink point of a vector field.	2
6	For what value of "a" is the vector $\vec{A} = (x+3y)\hat{\imath} + (y-2z)\hat{\jmath} + (x+az)\hat{k}$ solenoidal?	2

Part - B

Attempt any 2 questions

Each question carries 5 marks (5×2)

7.	Define curl of a vector point function. What is its physical significance?	2.5 + 2.5

8 (a) The wave function of a particle is given by $f(x) = cx$ for $0 \leq x \leq 1$. Find the probability that the particle lies between 0.35 and 0.75. **2.5 + 2.5**

(b) Consider the wave function $f(x,t) = A \exp(ax+ibt)$ where A, a and b are real positive numbers. Can it represent wave nature of a particle?

9. Write down differential equation of simple harmonic motion and solve it. **5**

Part - C

Attempt any 1 question

Each question carries 10 marks (10 × 1)

10. (a) Find the thickness of a wedge-shaped air film at a point where the fourth bright fringe is situated. Wavelength of sodium light is 5893 Å. **5+5**

(b) Prove that the horizontal motion of a spring-mass system on a frictionless surface is simple harmonic. Hence find the expression for its time period and frequency.

11 (a) Light of wavelength 5.9×10^{-7} m is incident on a thin soap film at 30° and dark bands are observed 5.0 mm apart. If $\mu = 1.33$, find the angle between the faces of the film. **5+1+4**

(b) What is the importance of doing normalization of wave function? Find the commutator relation between position and momentum.

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